

Advanced Apex Project-1 (M.Sc. DS S AI)

Week-5 Progress Submission

1. Project Title & Student Details

Customer Behaviour Profiling and Micro-Segmentation.

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2. Dataset Overview

UCI Online Retail II data set

3. Data Inspection & Audit

The retail transaction dataset contains **525,461 records** and **8 features** capturing invoice, product, customer, pricing, and geographic information. The data consists of both numerical variables (**Quantity** and **UnitPrice**) and categorical variables (**InvoiceNo**, **StockCode**, **Description**, **CustomerID**, and **Country**), providing a comprehensive view of customer purchasing behavior.

Preliminary data assessment reveals that **InvoiceDate** is currently stored as an object data type and should be converted to a datetime format to enable time-based analyses such as recency measurement, purchase frequency evaluation, and trend analysis. Missing values are primarily concentrated in **CustomerID (107,927 missing values)** and **Description (2,928 missing values)**, while all other variables are complete. The large number of missing customer identifiers requires careful handling, as customer-level analysis and segmentation depend on accurate customer identification.

The dataset includes **28,816 unique invoices**, **4,632 unique products**, **4,383 unique customers**, and transactions from **40 countries**, indicating substantial transactional diversity and a broad customer base. Repeated purchases by many customers suggest strong suitability for customer-centric analytical techniques such as RFM (Recency, Frequency, Monetary) analysis and behavioral segmentation.

As a real-world retail dataset, it is likely to contain cancellations, returns, duplicate transactions, and other forms of transactional noise that should be addressed during preprocessing. Overall, the combination of transactional, product, and customer information makes this dataset highly suitable for customer segmentation, behavioral profiling, spending pattern analysis, retention studies, and the development of personalized marketing strategies.

4. Data Preprocessing (DPP)

To ensure accurate and reliable customer behavior analysis, several data cleaning procedures were performed on the dataset. Records containing missing **CustomerID** values were removed because customer identification is essential for tracking purchasing behavior and performing customer-level segmentation. Transactions with missing **Description** values were also excluded to maintain consistency in product-related information.

In addition, transactions with **non-positive quantities** were removed, as these records typically represent returns, cancellations, or data entry errors rather than actual purchases. Similarly, transactions with **non-positive unit prices** were excluded because they do not reflect valid sales transactions and could distort spending and revenue calculations.

These preprocessing steps ensure that the final dataset contains only genuine and complete purchase transactions, providing a reliable foundation for feature engineering, RFM analysis, customer segmentation, behavioral profiling, and subsequent machine learning applications.

5. Feature Engineering

To support customer profiling and micro-segmentation, transaction-level data was aggregated into customer-level behavioral features. These engineered variables summarize purchasing patterns, spending behavior, and product preferences, making the data suitable for clustering and segmentation analysis.

i. Recency

Definition: Number of days since a customer's last purchase.

Formula:

Recency = Reference Date – Last Purchase Date

Interpretation: Lower values indicate recently active customers, while higher values may indicate inactive or churn-risk customers.

ii. Frequency

Definition: Total number of unique transactions made by a customer.

Formula:

Frequency = Count of Unique Invoices

Interpretation: Higher frequency reflects stronger customer engagement and loyalty.

iii. Monetary

Definition: Total revenue generated by a customer.

Formula:

Monetary = $\Sigma(\text{Quantity} \times \text{UnitPrice})$

Interpretation: Measures the overall financial value contributed by a customer.

iv. AvgBasketSize

Definition: Average number of items purchased per transaction.

Formula:

$$\text{AvgBasketSize} = \text{Total Quantity Purchased} / \text{Frequency}$$

Interpretation: Indicates the typical quantity of products purchased in each order.

v. ProductDiversity

Definition: Number of unique products purchased by a customer.

Formula:

$$\text{ProductDiversity} = \text{Count of Unique Product Descriptions}$$

Interpretation: Higher values indicate broader purchasing interests and greater product exploration.

vi. AvgOrderValue

Definition: Average spending per transaction.

Formula:

$$\text{AvgOrderValue} = \text{Monetary} / \text{Frequency}$$

Interpretation: Helps identify customers who make higher-value purchases.

Why These Features?

These features provide a comprehensive view of customer behavior:

- **Recency** measures customer freshness.
- **Frequency** measures engagement and loyalty.
- **Monetary** measures customer value.
- **AvgBasketSize** measures purchasing volume.
- **ProductDiversity** measures variety of interests.
- **AvgOrderValue** measures spending intensity.

Together, these metrics transform raw transaction data into meaningful customer profiles that can be used for clustering, customer segmentation, and targeted marketing strategies.

6. Exploratory Data Analysis (EDA)

i. Customer Behavior Distribution

Analysis of the engineered features reveals that customer behavior is highly uneven across the dataset. Most variables exhibit positively skewed distributions, indicating that the majority of customers have relatively low purchasing activity, while a small group demonstrates exceptionally high engagement and spending. Recency shows a moderate right skew, suggesting that most customers have purchased recently, whereas a smaller segment has remained inactive for longer periods.

ii. Spending and Engagement Patterns

Customer spending is concentrated among a relatively small subset of customers. While most customers contribute modest revenue, a few generate substantially higher sales, indicating the presence of high-value or VIP customers. Similarly, purchase frequency varies considerably across the customer base, with most customers making only a limited number of purchases and a small number exhibiting very high transaction counts. These highly engaged customers are likely to represent important targets for retention and loyalty initiatives.

iii. Product Diversity

Product diversity differs significantly across customers. Most customers purchase a limited range of products, while a smaller group explores a much broader selection of the product catalog. Customers with greater product diversity generally appear to be more engaged and active overall.

iv. Correlation Analysis

Correlation analysis reveals several meaningful relationships among the behavioral metrics. Frequency and ProductDiversity show the strongest positive relationship, indicating that customers who purchase more often tend to buy a wider variety of products. Frequency also demonstrates a strong positive correlation with Monetary value, suggesting that frequent purchasers typically generate higher revenue. Monetary value and ProductDiversity exhibit a moderate positive relationship, while Recency is negatively correlated with both Frequency and ProductDiversity, implying that recently active customers tend to purchase more often and explore more products.

v. Outlier Analysis

Several features, particularly Frequency, Monetary, AvgBasketSize, ProductDiversity, and AvgOrderValue, contain extreme values. These observations are likely to represent genuine high-value customers rather than data quality issues. Since clustering algorithms such as K-Means are sensitive to extreme values and feature scale, appropriate feature scaling is necessary before model development.

vi. Pairwise Relationships

Pairwise visualizations further confirm a positive relationship between purchase frequency and spending behavior. Customers who purchase more frequently generally contribute higher monetary value. The visual separation of a small group of customers from the majority also suggests the existence of naturally occurring customer segments.

vii. Data Quality and Segmentation Potential

The engineered dataset contains no missing values, indicating that the preprocessing and feature engineering stages successfully produced a complete customer-level dataset. The substantial variation observed across customer behavior metrics highlights the diversity of the customer base and suggests strong potential for meaningful segmentation. Overall, the dataset is well-suited for unsupervised learning techniques such as K-Means clustering to identify distinct customer groups and support targeted marketing strategies.

7. Preliminary Observations & Learnings

The exploratory analysis provides valuable insights into customer purchasing behavior and confirms that the dataset is well-prepared for customer segmentation.

i. Data Quality and Preparation

The dataset was successfully cleaned by removing records with missing customer identifiers, incomplete product descriptions, and invalid transactions containing non-positive quantities or prices. As a result, the final dataset accurately represents genuine customer purchases and provides a reliable foundation for customer-level analysis.

ii. Customer Behavior Patterns

Customer behavior is highly heterogeneous across the dataset. While most customers make relatively few purchases and contribute modest revenue, a small group demonstrates exceptionally high purchasing activity and spending. This imbalance suggests the presence of distinct customer groups with varying levels of engagement and value.

iii. Spending and Product Engagement

Revenue is concentrated among a limited number of customers, indicating the existence of high-value or premium customer segments. Similarly, customers vary considerably in the range of products they purchase. More active customers generally explore a broader selection of products, suggesting that product diversity is closely associated with overall engagement.

iv. Relationships Between Features

Strong positive relationships were observed between Frequency and ProductDiversity, as well as between Frequency and Monetary value. These findings indicate that customers who purchase more frequently tend to spend more and buy a wider variety of products. In contrast, Recency shows negative relationships with both Frequency and ProductDiversity, suggesting that recently active customers are generally more engaged and involved with the business.

v. Outliers and Customer Value

Several behavioral metrics contain significant outliers, particularly in spending, purchase frequency, basket size, and product diversity. These outliers are likely to represent genuine high-value customers rather than data errors. Their presence highlights the need for appropriate feature scaling and transformation before applying clustering algorithms.

vi. Readiness for Customer Segmentation

The engineered feature set contains no missing values and effectively captures key dimensions of customer behavior, including recency, loyalty, spending power, purchasing volume, and product exploration. The substantial variation observed across these metrics indicates strong potential for identifying meaningful customer segments.

Key Learning

The analysis demonstrates that the customer base is not homogeneous. Customers differ significantly in their purchasing frequency, spending habits, order sizes, and product preferences. These behavioral differences provide a strong basis for clustering customers into distinct segments, enabling more targeted marketing strategies, improved customer retention efforts, and enhanced business decision-making.

8. Planned Next Steps

The exploratory analysis confirms substantial variation in customer purchasing behavior, spending patterns, and product engagement. While these findings provide valuable insights, further analysis is required to transform them into actionable business strategies. Therefore, the next phase of the project will focus on customer segmentation and cluster profiling.

i. Customer Behavior Profiling

The engineered customer-level features will be prepared for clustering by addressing the strong skewness observed in several variables and applying appropriate feature scaling techniques. Standardization will ensure that all features contribute equally to the clustering process and prevent variables with larger numerical ranges from dominating the analysis.

Objective: Develop a robust and representative customer dataset suitable for unsupervised learning and segmentation.

ii. Customer Micro-Segmentation

K-Means clustering will be applied to group customers with similar behavioral characteristics. Multiple cluster configurations will be evaluated to identify the most meaningful segmentation structure. Each customer will then be assigned to a specific segment based on their purchasing behavior, spending patterns, and engagement levels.

Objective: Identify naturally occurring customer groups and uncover hidden behavioral patterns within the customer base.

iii. Cluster Evaluation and Profiling

The quality of the clustering results will be assessed using techniques such as the Elbow Method and Silhouette Score. Once an optimal number of clusters has been identified, each segment will be profiled by analyzing its characteristics across Recency, Frequency, Monetary Value, Product Diversity, Average Basket Size, and Average Order Value.

Objective: Validate the effectiveness of the segmentation model and understand the defining traits of each customer group.

iv. Business Insights and Strategic Recommendations

The identified customer segments will be visualized and interpreted from a business perspective. Particular attention will be given to recognizing high-value customers, loyal customers, low-engagement customers, and customers at risk of churn. The segmentation results will then be translated into actionable recommendations to support business growth and customer relationship management.

Potential applications include:

- Customer retention and reactivation strategies
- Loyalty program optimization
- Personalized marketing campaigns
- Cross-selling and upselling opportunities
- Revenue and profitability improvement initiatives

Objective: Convert analytical findings into practical business actions that enhance customer engagement and long-term value.

Expected Outcome

By the completion of the project, customers will be segmented into meaningful behavioral groups based on their purchasing activity, spending behavior, and product engagement. These segments will provide a deeper understanding of customer value and preferences, enabling more targeted marketing efforts, improved customer retention, and data-driven business decision-making. Ultimately, the segmentation framework will help the organization better understand its customer base and develop strategies that drive sustainable growth and customer satisfaction.